

7-MODULE MAP

Blended consultant programme · seven weeks, one module per week · every module ends in a working delivery example.

AI marks where AI is used in the work

MODULE 1 FOUNDATION LAYER FOUNDATIONS & SPEC-DRIVEN DELIVERY

Reproducible environments & tooling

Isolated Python with linting, typing and tests, the baseline for verifying generated code.

Git & code review (deepened)

Work fluently across the Git model, with peer review as a first-class skill.

The engineering bar for the AI layer

Hold the AI layer to the same test, type and review standard as the rest of the stack.

Spec-driven development, the spine **AI**

Deliver from a precise spec: intent, then plan, then code, with a review between.

The AI pair-programming loop **AI**

Run the loop and own every line: scope, plan, review the diff, verify.

MODULE 4 APPLICATION LAYER RETRIEVAL QUALITY & EVALS

RAG failure modes **AI**

Provoke and mitigate ungrounded hallucination, lost-in-the-middle and citation drift.

Chunking strategy, judged by metrics **AI**

Select a strategy backed by recall@k and answer quality rather than intuition.

Hybrid retrieval **AI**

Combine lexical BM25 and vector retrieval with rank fusion, diagnosing each win.

Reranking & query rewriting **AI**

Measured against latency and cost, to prove the retrieval lift is real.

Evals & LLM-as-judge, audited **AI**

Design a valid eval set, then audit the judge for bias and variance.

MODULE 7 DELIVERY / PROJECT LAYER CLIENT DELIVERY SPRINT & DEFENCE

Discovery & scoping (deepened)

Turn a vague problem into a scoped, shippable increment with an assumption log.

Pragmatic architecture

Choose boring, reliable architecture under pressure and reuse prior building blocks.

Walking-skeleton delivery **AI**

Direct agents to the thinnest end-to-end path first, with a deliberate debt log.

Absorbing change & client communication

Re-scope and negotiate injected change while protecting the demo target.

The reuse artefact & defence **AI**

Document the reusable component, then defend correctness, readiness and reusability.

MODULE 2 FOUNDATION LAYER APIS, SERVICES & THE LLM IN ANGER

APIs from both sides

Build and consume REST APIs with sound handling of auth, pagination and rate limits.

Robust services

Idempotency, retries with backoff, timeouts and FastAPI, with the contract owned.

The first LLM call, robustly **AI**

Retries, backoff and timeouts, reading token usage and stop reasons.

The LLM API in anger, verified **AI**

Cost-aware calls with streaming and structured output, catching hallucinated APIs.

LLM behaviour & context engineering **AI**

Control behaviour through context: tokens, window, temperature and roles.

MODULE 5 APPLICATION LAYER AGENTS, ORCHESTRATION & GUARDRAILS

Agentic patterns & routing **AI**

Route with confidence and fallback, using the least powerful pattern that works.

Single & multi-agent orchestration **AI**

State graphs, tool nodes and sub-graphs, with deliberately designed hand-offs.

Retrieval as a tool **AI**

Wire grounded, cited retrieval as a tool and evaluate retrieval, not just the answer.

Connectors, MCP & guardrails **AI**

Connect tools via MCP behind input validation, allow-lists, step caps and HITL.

Emergent failure & observability **AI**

Capped iterations, deterministic fallbacks, spans, and prompt-injection defence.

SEVEN WEEKS, SEVEN MODULES

WEEK 1	Foundations & spec-driven delivery
WEEK 2	APIs, robust services, the LLM in anger
WEEK 3	Context engineering & first RAG
WEEK 4	Retrieval quality & evals
WEEK 5	Agents, orchestration & guardrails
WEEK 6	Production hardening, cost & platform
WEEK 7	Client delivery sprint & defence

HOW EVERY WEEK ENDS

Ship Demo Day: working software demonstrated every Friday and explained line by line. No demo without a talk-through, and from Week 4 no quality claim without numbers from the learner's own eval harness.

WHAT LEAVES WITH THEM

A delivered client increment, an audited eval set, a guarded agent with an audit trail, a versioned cost-bounded endpoint, and the reusable component the next engagement starts from.

MODULE 3 APPLICATION LAYER CONTEXT ENGINEERING & FIRST RAG

Embeddings & where they fail **AI**

Build similarity intuition and locate the breaks: negation, exact IDs, rare terms.

The RAG pipeline, scaffolded & verified **AI**

Stand up retrieve-augment-generate, then verify chunking, indexing and retrieval.

Grounding, citations & refusal **AI**

Grounded answers with sources, and a refusal when no supporting context exists.

Structured output & tool use **AI**

JSON-validated output and a working tool-use loop with repair-and-retry.

Rapid UI prototyping **AI**

Usable streaming prototypes with Streamlit, Gradio and AI-generated React.

MODULE 6 PRODUCTION LAYER PRODUCTION HARDENING & PLATFORM

Prototype to service

Environment config, structured logging, tests and a health endpoint.

Deploy & versioning

Managed endpoint with traffic splitting, canary releases and autoscaling.

Cost & latency, measured

Caching, right-sizing, batching and quantisation, benchmarked at p50/p99.

Monitoring & drift

Operational, quality, drift and safety axes, thresholds tied to real failure modes.

Prompts as code & platformise **AI**

Version prompts with golden tests, then extract a parametrised reusable core.